Session 18 Quiz - AI in Everyday Life & Across Industries

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57497]
Fill in the blank: The three-step 'recipe' used by every AI system is Sense, ________, and Act.

A. Speak
B. Learn
C. Watch
D. Think

Correct Answer: D. Think

Explanation: The middle step is "Think", where the AI processes information to make a decision. Speak, Learn, and Watch are not the standard term for this processing stage.

Q2 [Type: MCQ | Difficulty: Easy | QID:57498]
Fill in the blank: When an AI analyzes the pixels in a medical scan to find signs of illness, it is using a technology called Computer ________.

A. Logic
B. Science
C. Vision
D. Programming

Correct Answer: C. Vision

Explanation: Computer Vision allows AI to "see" and process visual information, such as medical images. Logic, Science, and Programming are not the correct terms for this visual analysis technology.

Q3 [Type: MCQ | Difficulty: Easy-Medium | QID:57499]
Fill in the blank: The ________ rule ensures that humans remain responsible for making the final decisions that affect people's lives.

A. AI-First
B. Digital-only
C. Automated-Action
D. Human-in-the-loop

Correct Answer: D. Human-in-the-loop

Explanation: This principle ensures that AI serves as a helper while a person retains final control over critical outcomes. AI-First, Digital-only, and Automated-Action do not emphasize human responsibility.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57500]
Fill in the blank: In the finance industry, AI watches for unusual spending patterns to detect ________ activity instantly.

A. Shopping
B. Fraudulent
C. Standard
D. Marketing

Correct Answer: B. Fraudulent

Explanation: AI identifies transactions that do not match a user's normal habits to prevent theft or unauthorized use (fraud). Shopping, Standard, and Marketing are not the types of activity AI detects in this context.

Q5 [Type: MCQ | Difficulty: Easy | QID:57501]
Fill in the blank: In the ________ industry, NASA uses AI to analyze data from rovers to find new planets and stars.

A. Navigation
B. Space
C. Atmospheric
D. Aviation

Correct Answer: B. Space

Explanation: Space exploration involves massive amounts of data that AI can process faster than humans to find celestial bodies. Navigation, Atmospheric, and Aviation are not the primary industries for NASA's exoplanet discovery.

Q6 [Type: True/False | Difficulty: Easy-Medium | QID:57502]
True or False: AI is intended to completely replace teachers by grading essays and generating practice questions.

A. False
B. True

Correct Answer: A. False

Explanation: AI is described as a tool that helps teachers, rather than a replacement for the human connection in education. The statement says "completely replace teachers," which is false.

Q7 [Type: True/False | Difficulty: Easy | QID:57503]
True or False: Most of a commercial flight is managed by AI autopilot, which handles speed, navigation, and altitude.

A. False
B. True

Correct Answer: B. True

Explanation: Autopilot systems are sophisticated enough to manage the majority of a flight's technical operations under human supervision. This is true for modern commercial aviation.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57504]
True or False: In the entertainment industry, real humans are still the creators because AI is used only as a tool to help tell their stories.

A. True
B. False

Correct Answer: A. True

Explanation: AI can generate art or effects, but the core storytelling and creative vision come from human artists. Humans remain the creators, with AI as a tool.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57505]
True or False: Smart traffic lights use AI to change signals based on how many cars are waiting at an intersection.

A. False
B. True

Correct Answer: B. True

Explanation: This application of AI reduces congestion by sensing real-time traffic flow rather than using a fixed timer. Smart traffic lights do adapt to car volume.

Q10 [Type: True/False | Difficulty: Easy | QID:57506]
True or False: AI is only used in digital apps on smartphones and has no impact on physical industries like farming or climate tracking.

A. True
B. False

Correct Answer: B. False

Explanation: AI is used across many real-world industries beyond phone apps, including agriculture (drones for crop monitoring) and climate science (tracking animal migration and ice melt). The statement is false.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57507]
Which of the following is an example of the 'Think' step in a YouTube recommendation system?

A. Rerouting you to a faster internet server
B. Calculating that you are likely to enjoy more gaming content
C. Recording that you clicked on a 'gaming' video
D. Displaying a video thumbnail on your screen

Correct Answer: B. Calculating that you are likely to enjoy more gaming content

Explanation: This involves processing data to make a prediction, which is the core of the Think phase. Recording clicks is Sense, displaying thumbnails is Act, and rerouting is networking, not AI thinking.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57509]
In the healthcare industry, how does AI primarily assist nurses when monitoring patients?

A. By cleaning the hospital rooms
B. By performing surgery automatically
C. By alerting staff if a heart rate drops significantly
D. By deciding which patient gets to go home first

Correct Answer: C. By alerting staff if a heart rate drops significantly

Explanation: AI can continuously monitor vitals and act by sounding an alarm for human medical professionals. It does not clean rooms, perform surgery, or make discharge decisions autonomously.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57510]
What is the primary role of AI in the farming industry according to the source material?

A. Selling the harvested food online
B. Using drones to check crops and predict planting times
C. Replacing all human farm workers
D. Designing new types of vegetables

Correct Answer: B. Using drones to check crops and predict planting times

Explanation: AI helps farmers make data-driven decisions by analyzing images of fields and weather patterns. It does not sell food, design vegetables, or replace all workers.

Q14 [Type: MCQ | Difficulty: Medium | QID:57512]
Why is the 'Human-in-the-Loop' model critical for banking and finance AI?

A. Because AI cannot see credit card numbers
B. Because humans are faster at math than computers
C. To manually count all the money in the bank
D. To ensure a person confirms fraud before an account is locked

Correct Answer: D. To ensure a person confirms fraud before an account is locked

Explanation: This prevents AI from accidentally locking someone out of their money due to a mistake or unusual but valid purchase. Human confirmation is essential for high-stakes decisions.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:57513]
Which of these is an everyday example of AI using 'Natural Language Processing' (NLP)?

A. Autocorrect and predicted text
B. Heart rate monitoring
C. Facial Recognition
D. GPS traffic rerouting

Correct Answer: A. Autocorrect and predicted text

Explanation: NLP allows AI to understand, break down, and generate human language to fix spelling or suggest words. Heart rate monitoring is sensor data, facial recognition is computer vision, and GPS routing uses location data.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57514]
How does AI provide 'personalized' education for students?

A. By requiring students to use tablets for every task
B. By adapting practice questions based on a student's weak spots
C. By replacing human teachers with robots
D. By teaching every student the same lesson at the same time

Correct Answer: B. By adapting practice questions based on a student's weak spots

Explanation: AI can "Sense" wrong answers, "Think" about what a student struggles with, and "Act" by providing specific help. This is personalized learning, not uniformity or replacement.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57515]
In a self-driving car, which action represents the 'Sense' part of the AI recipe?

A. Applying the brakes to stop at a red light
B. Calculating the safest route home
C. Using cameras and radar to detect pedestrians
D. Turning the steering wheel

Correct Answer: C. Using cameras and radar to detect pedestrians

Explanation: Sensing involves gathering data from the environment through hardware like cameras or sensors. Braking and steering are Act, calculating a route is Think.

Q18 [Type: MCQ | Difficulty: Easy | QID:57516]
Which industry uses AI to track melting ice and animal migration patterns?

A. Education
B. Transportation
C. Climate
D. Entertainment

Correct Answer: C. Climate

Explanation: AI helps scientists process massive amounts of satellite and sensor data to understand changes in our environment. This is climate science, not education, transportation, or entertainment.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57517]
What does Google Maps 'Think' about when suggesting a new route during your ride?

A. What music you are currently listening to
B. The history of your phone's battery life
C. The color of your car
D. The fastest path based on real-time traffic data

Correct Answer: D. The fastest path based on real-time traffic data

Explanation: The algorithm processes traffic speeds from millions of drivers to find the most efficient route. Music, battery life, and car color are not relevant to route planning.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57518]
Which of these AI moments would be considered 'low-risk' if the AI made a mistake?

A. A YouTube video recommendation you didn't like
B. An AI deciding who goes to jail
C. A self-driving car misidentifying a stop sign
D. An AI deciding who receives a heart transplant

Correct Answer: A. A YouTube video recommendation you didn't like

Explanation: This is low-risk because the only consequence is that you spend a few seconds finding a different video. The other options involve life, liberty, or physical harm, which are high-risk.